

Summary of Workspace Topics

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Based on the contents and structure of the workspace, the following major research topics and computational experiments have been explored:

- **Ground Penetrating Radar (GPR) Simulation & Modeling:** Building forward models using gprMax, incorporating complex geometries like subwavelength scatterers, thin beds, and baseline structural models.
- **Full Waveform Inversion (FWI):** Developing FWI workflows (using tools such as Devito), experimenting with objective functions including quantum/geometric misfits, and employing optimization techniques like L-BFGS.
- **Time-Reversal & Array Signal Processing:** Implementing high-resolution imaging algorithms, specifically Time-Reversal MUSIC (TR-MUSIC), to analyze diffractions and localize small targets.
- **Reverse Time Migration (RTM):** Processing synthetic B-scan data to produce migrated subsurface images.
- **Wave Scattering & Diffraction Analysis:** Conducting focused experiments on horizontal scattering phenomena, multi-transmitter setups, and subwavelength resolution limits.
- **Bayesian Inference & Uncertainty Quantification:** Applying Markov Chain Monte Carlo (MCMC) methods for probabilistic inversion and parameter estimation.